

ONKAR KOTULE

Software Development Engineer (SDE-2) | Backend & Full-Stack | MERN | Azure | Mendix

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PROFESSIONAL SUMMARY

Software Development Engineer with **4+ years of experience** building scalable backend systems and full-stack applications across enterprise and startup environments. Strong expertise in **API design, data modeling, and performance optimization**, with hands-on experience designing **transaction-safe and real-time systems**. Proficient in developing **high-performance, production-grade architectures** using Node.js, MongoDB, and cloud platforms, with a focus on **scalability, consistency**, and clean system design. Proven ability to **own features end-to-end** and build reliable, scalable systems in production.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Java, Python.

Frontend: React.js, Next.js, React Native, Expo, Tailwind CSS, Redux Toolkit, TanStack Query.

Backend: Node.js, Express.js, MongoDB, PostgreSQL, REST APIs, GraphQL, JWT, OAuth.

Cloud & DevOps: Azure DevOps, CI/CD, Docker, Git, GitHub, Firebase, Appwrite.

Testing: Jest, Basic Unit Testing.

AI/GenAI: langGraph, LLM Integration, Prompt Engineering, Agentic Workflows.

PROFESSIONAL EXPERIENCE

Associate Consultant (Software Engineer – Backend & Full-Stack)

Capgemini – Pune, India | Jul 2023 – Present

- Designed and developed enterprise-scale applications across backend and frontend, optimizing **state management, data flow, and UI performance for complex workflows**.
- Architected backend integrations and **REST API layers** for scalable communication across services
- Built systems with **role-based access control (RBAC), auditability, and secure data flows**.
- Owned features end-to-end, from **requirement design and development** to production **deployment and post-release ownership**.

Key Contributions:

- Designed a **document management system (ShareSpace)**:
 - Hierarchical file storage with **optimized data modeling**
 - Secure upload/download pipelines with **access control**
 - Soft delete (trash), restore & lifecycle management
- Refactored **Export Control system**:
 - Converted rigid logic into **config-driven architecture**
 - Improved maintainability and reduced manual effort
- AI-Powered **Investment Insights System**:
 - Designed **LLM-based pipelines** to generate structured investment reports
 - Implemented **prompt engineering and response validation** for consistent and reliable outputs
 - Built **dynamic frontend components** to render UI based on API-driven responses
 - Integrated **AI-generated insights into applications** for real-time user consumption

- Built **Dangerous Goods Transport System**:
 - Slot booking system with **workflow orchestration**
 - Real-time notifications and approval pipelines
 - Implemented:
 - **Azure AD SSO integration**
 - CI/CD pipelines with Azure DevOps
 - Production monitoring & issue resolution (SPOC role)
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Software Developer (Backend & Full-Stack)

BITCS | DigitalFlow | KnockOnce | Jun 2021 – Oct 2022

- Built scalable backend services using Node.js, Express, and MongoDB, designing modular REST APIs with validation and error handling.
- Developed frontend systems using **Next.js and Redux Toolkit**, implementing optimized state management and efficient data fetching.
- Optimized frontend performance using:
 - Efficient **state management and component reusability** (React, Redux Toolkit).
 - Data **fetching and caching** strategies (TanStack Query).
 - **Reduced unnecessary re-renders** and improved **UI responsiveness**.
- Improved system performance through **MongoDB indexing, query optimization, and efficient schema design**

Expense management system

- Designed and developed a **production-grade expense management system** handling multi-user financial transactions
- Implemented:
 - **Transaction-safe balance updates** ensuring data consistency across users
 - **Real-time expense tracking** and settlement logic
- Designed optimized data models and aggregation pipelines for **analytics and reporting**
- Enhanced performance using **compound indexes for high-frequency queries**
- Built the system with focus on **scalability, concurrency handling, and data integrity**

Additional Systems Built

- Built real-time communication features using **event-driven architecture (Socket.IO)**
 - Developed modular frontend systems with optimized data fetching & caching
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CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate (AZ-204)
 - Microsoft Certified: Azure AI Engineer Associate (AI-104)
 - Microsoft GenAI Practitioner (AI-900)
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EDUCATION

Bachelor of Engineering (B.E.)

Bharati Vidyapeeth College of Engineering, Pune

